

Electronic Vehicle Battery Disassembly Using AgenticAI specifications

Abstract—

I. INTRODUCTION

The continuous adoption of electronic vehicles (EV) poses a growing challenge in recycling their batteries [1], which contain a series of high-value materials that can often be recuperated to a large percentage in excess of 90% [2]. With battery voltages now reaching 800V and a significant risk of a difficult-to-extinguish self-incineration due to thermal runaway effects, manual disassembly is not only dull and dirty, but also dangerous. At the same time, economic incentives are dwindling: whereas current batteries are made of lithium, nickel, and cobalt (LNC), the new generation of mass-market batteries is based on lithium iron-phosphate (LFP), with raw materials costs of less than a quarter. This increases pressure on the industry to find solutions that reduce the cost of disassembly. One possible solution is to shred the entire battery. The resulting “black mass” is of lower quality, and therefore has a significantly lower price, than black mass that is obtained from shredding only the battery cells themselves [3]. Shredding the entire battery, including aluminum framing, copper busbars, and steel screws, also creates significant wear on the shredder gears that takes away from the unit economics of recycling. This has created considerable interest in robotic dismantling of EV battery cells. An emerging technique is to cut the battery using lasers [4]. This is particularly important for so-called cell-to-pack architectures in which cells are not grouped into individual modules, but glued together into a large monolithic structure, as well as designs that extensively rely on welding of busbars instead of reversible connections. We consider cutting techniques to be complementary to robotic disassembly, in particular if reuse of components for remanufacturing is desired.

Complete autonomy requires complex robotic systems that need to perform a large variety of tasks including screwing, cutting, prying, etc. In [5], a complete disassembly system for a hybrid battery is described and evaluated at the component disassembly level. Such systems usually require a large number of specialized tools, the ability to tool change, and specialized end-effectors to remove parts from the disassembly. For example, in [6] a tool for automatically opening snap-fit connectors is presented and in [7] a tool for removing screws after unscrewing is described. In [8], four robot arms, each with dedicated tools, are used for complete disassembly of a plug-in hybrid battery. Here, one robot can swap nuts on the nutrunner operated by the other robot. This work also directly compares dismantling times with that of



Fig. 1. Hyundai Ioniq5 EV battery is being dismantled by a custom nutrunner mounted on a UR16e collaborative robot and a 2.1m Parker linear axis.

a human operator, noting that robotic operations tend to be 20-30% slower.

In order to reduce the complexity of the robotic solution, others [9], [10] propose a human-robot collaborative approach, using the Audi Q5 hybrid battery pack as an example, in which the robot is focused on repetitive screwing tasks, whereas the human is performing tasks that are more difficult to automate.

Due to the large variety of tasks and battery packs, many of which require significant dexterity, we believe a human-robot collaborative approach to persist in industry and advocate a solution that can incrementally augment human assembly stations.

A. Related work

We note that the majority of the related work is on the much smaller category of hybrid battery packs (usually small enough to fit in the area underneath the trunk) [5], [9], [10], [8], not on the emerging class of full-size EV battery packs that often span the entire chassis of the car, and operate at lower voltages (48V vs. 400-800V). These larger batteries exceed the workspace even of large industrial arms. The disassembly problem of *packs* that consists of multiple *modules*, should also not be confused with the disassembly of individual modules into *cells* [11].

In addition to performing specific tasks of the disassembly process [6], [7], robotic disassembly taps into multiple areas of robotics such as task and motion planning. [12] investigates a task planning framework for the complete disassembly of an Audi e-tron hybrid battery pack. Here, the planner

is hard coding disassembly logic such as removing all visible screws before attempting the next step. In [13] a hierarchical planner is employed to disassemble a wooden toybox and an electronic hard-drive with emphasis on human-robot collaboration. In our work, we use Large Language Models (LLM) to interactively create a plan together with a human user. Here, LLMs can leverage common-sense knowledge about disassembly and also directly issue high-level commands to a robot's sensors and actuators [14]. In particular, LLMs can be used interactively to let a user refine a robotic manipulation plan [15]. In [16], an Agentic AI framework has been presented that facilitates interactions between an LLM and sensors and actuators controlled by the Robot Operating System (ROS2). In addition to using the LLM to generate high-level plans, they also show using the LLM to detect anomalies during the execution of classical Finite State Machines (FSM), thereby potentially making robots more robust.

Finally, [17] presents a techno-economic analysis of the battery disassembly process, computing cost savings larger than 40% when using one Universal Robot arm to support dismantling of the Mitsubishi Outlander hybrid battery.

B. Contributions

This paper presents a human-robot collaborative disassembly solution for a *full-size* EV battery consisting of a robotic arm and overhead gantry system, as well as an agentic large language model approach to program and plan the disassembly system as well as collaborate with a human co-worker. s

II. SYSTEM OVERVIEW

The dismantling system consist of a Universal Robot UR16e, a customized nut-runner (AlloyPower with custom Ethernet interface), and a Parker linear gantry. Figure ?? shows the lab prototype with a Hyundai Ioniq5 full-size EV battery. The end-effector is equipped with an Intel Realsense D435. Images are processed via Yolo World and motion planning is performed using MoveIt 2! [18]. The overall operation is orchestrated using an agentic AI framework SmolAgents [19]. Figure 2 shows an overview of the system.

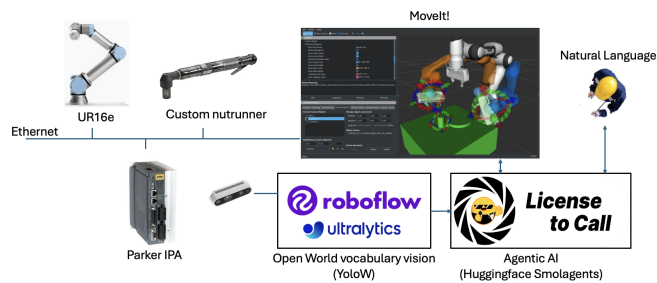


Fig. 2. Integration of conventional automation systems, open-world vision systems, motion planning and Agentic AI.

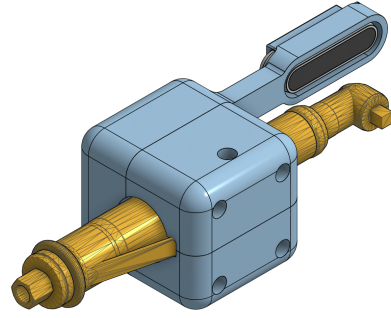


Fig. 3. Custom end-effector for the UR16e that holds a nut-runner and an Intel Realsense D435 in place.

A. Electric Vehicle Battery: Hyundai Ioniq5

The Hyundai Ioniq5 is a full-size LNC EV battery that is 2.12m in length, 1.22m wide, weighs approximately 450kg, has a capacity of 77.4kWh, and operates at 800V. We have chosen this battery as it can be reversibly assembled and disassembled. Compared with hybrid or plug-in batteries, this kind of batteries require a much larger workspace, and additional safety precautions due to the high voltage, but has significantly different techno-economic benefits due to the increased amount of high-value materials.

We have performed a complete manual disassembly of the battery to obtain an estimate of manual disassembly time using two untrained workers and a third person for time keeping who assisted in some of the tasks. We report and sum up the time required for all individual steps, not including breaks, and normalize them by “man-hour” and “man-seconds”. Times were recorded after performing one complete disassembly and reassembly with the same team, somewhat reducing the effect of learning. Prior to disassembly, we have discharged the battery using an Webasto/Aeronvironment AV-900 Heavy Duty Dual Channel Cycling battery cycler to 0V.

We identified 12 individual subtasks, which are listed in table I.

Table II lists all of the components and their estimated recycling value.

B. Robotic System

The robotic system has been dimensioned to service full-size EV batteries and consists of a T-slot arch spanning 2.5m in width, a Parker-Hannifin linear axis (2.1m) and an Universal Robots 16e 6-DoF robotic arm with a payload of 16 kg. Both the Intel RealSense D435 and the nut-runner are mounted using a custom, 3D-printed assembly shown in Figure 3. Here, the camera is mounted slightly offset so that the nut-runner does not interfere with its field of view. The nut-runner is mounted so that its torque axis is perpendicular to the last axis of the robot, preventing reaction forces to be absorbed by the wrist.

The UR16e provides a socket interface to send URScript commands. We have created custom scripts that allow us

TABLE I
BATTERY DISASSEMBLY TASK BREAKDOWN

Task	Description	Time (s)	Workers	Man-seconds	Tools
1	Remove 8 large screws on top cover	120	2	240	Large nutrunner
2	Remove 70 screws and nuts holding top cover	501	2	1002	Nutrunner (high torque)
3	Remove cover plate	27	2	54	None
4	Remove two busbars to break battery voltage into two	100	2	200	Nutrunner (high torque)
5	Remove two large cables connecting across battery	173	2	346	Nutrunner (high torque)
6	Remove 6+1 large busbars	190	2	380	Nutrunner (high torque)
7	Remove 3 busbars from battery controller	136	2	272	Nutrunner (high torque)
8	Remove 8 wire harnesses	420	2	840	Small screwdriver, flathead
9	Take out 24 small bus bars (battery module interconnects)	456	2	912	Nutrunner (high torque), small screwdriver
10	Take out first layer of screws holding in battery modules (80 screws)	960	2	1920	Nutrunner (high torque)
11	Take out second layer of M14 screws	307	2	614	Nutrunner (high torque) M14
12	Remove 32 battery modules	343	3	1029	Crowbar

TABLE II
BATTERY COMPONENT WEIGHT AND MATERIAL VALUES

Part	Weight	Quantity	Total	Material	Price/kg	Total Value
Module	11732g	32	375.42kg	NCM Blackmass	\$3.50	\$1,313.98
Busbar S	483g	6	2.898kg	Copper	7.06401766	\$20.47
M10 short/beefy	14		0	Steel	0.15	\$-
M10 long	12		0	Steel	0.15	\$-
M10 short	6		0	Steel	0.15	\$-
Small bus bars	36	24	0.864	Copper	7.06401766	\$6.10
M14	13		0	Steel	0.15	\$-
Top cover screws (large)	109	8	0.872	Steel	0.15	\$0.13
Top cover screws (small)	3	77		Steel	0.15	\$-
Wire harness	143g	8	1.144	Copper/insulation	2.538631347	\$2.90
Top cover nuts		78	0	Steel		\$-
Other bus bars	500g	1	0.5	Copper/insulation		\$-
Aluminium bottom	60kg	1	60kg	Aluminium	1.103752759	\$66.23
Top cover	3kg	1	3kg	Steel	0.15	\$0.45

to move the robot toward a screw location in force control mode. Once the robot makes contact, the nut-runner is activated. As the nut-runner removes the screw, the robot automatically moves upward as it maintains a constant downward force.

The nut-runner is equipped with a button that allows to regulate the nut-runner speed by modulating a pulse-width modulated (PWM) signal. We have replicated the built-in circuitry in order to control the nut-runner via an Ethernet connection. The schematic is shown in Figure XX.

The gantry system can be controlled by sending assembler-like commands via another socket connection. It is equipped with an absolute encoder, which allows to move it with millimeter accuracy.

The overall system is controlled by MoveIt! [18], which implements path planning and collision avoidance. To this end, we are scanning the entire battery using the Intel Realsense to create a point-cloud collision object.

III. AUTONOMOUS DISASSEMBLY

We envision the following industrial workflows: A preliminary setup sequence that will need to be done only once for every battery type

- 1) Scanning the battery,
- 2) Manually labeling all relevant parts,
- 3) Training a vision model,

- 4) Map all relevant parts using vision and store their location in a database,
- 5) Manually annotate missing parts,
- 6) Create disassembly plans by providing text input, and the actual disassembly sequence,

- 1) Scan the entire battery and create a collision object,
- 2) Use vision to register the battery pose,
- 3) Execute the disassembly plan by issuing commands to both the robot and the worker,
- 4) Wait for worker input to continue the disassembly plan.

We describe the details of the individual steps in the sections below.

A. Planning

We formulate the sequencing of disassembly tasks as a Traveling Salesman Problem (TSP) defined over task poses in $SE(3)$. Let

$$T_s = \{t_{s,1}, \dots, t_{s,n}\}$$

denote the set of disassembly tasks that are associated with step $s \in \mathcal{S}$, and $t_{s,0}$ represents the robot home configuration. Each task pose is defined as

$$t_{s,i} = (p_{s,i}, q_{s,i}) \in SE(3),$$

where $p_{s,i} \in \mathbb{R}^3$ is the Cartesian position and $q_{s,i} \in \mathbb{H}$ is a unit quaternion representing orientation. The augmented

node set is

$$V_s = \{t_{s,0}\} \cup T.$$

Rather than defining edge costs through a geometric distance in task space, which does not reflect planning constraints, we construct a weight matrix

$$W_{s,ij} \in \mathbb{R}_{\geq 0}, \quad i, j \in V_s,$$

where each entry represents the minimum joint-space motion cost between task poses $t_{s,i}$ and $t_{s,j}$. Specifically, let

$$\mathcal{Q}_{s,ij} = \{q_{s,ij}^{(1)}, \dots, q_{s,ij}^{(N)}\}$$

denote a set of N inverse kinematic solutions for reaching pose $t_{s,j}$ from pose $t_{s,i}$, obtained via RRT* in joint space. The edge weight is then defined as

$$W_{s,ij} = \min_{k \in \{1, \dots, N\}} c(q_{s,ij}^{(k)})$$

where $c(\cdot)$ denotes the joint-space path cost (e.g., path length, time, or energy). Note that $W_{s,ij}$ can be computed offline as the set of tasks T is the same for each battery of the same type.

The TSP decision variables are

$$x_{s,ij} \in \{0, 1\}, \quad i, j \in V,$$

with $x_{s,ij} = 1$ if the robot moves directly from $t_{s,i}$ to $t_{s,j}$. The optimization problem is

$$\min_{x_{s,ij}} \sum_{i \in V} \sum_{j \in V} W_{s,ij} x_{s,ij},$$

subject to the degree constraints

$$\sum_{j \in V_s} x_{s,ij} = 1, \quad \sum_{i \in V_s} x_{s,ij} = 1 \quad \forall i, j \in V,$$

and standard subtour elimination constraints ensuring a single Hamiltonian circuit. This formulation decouples global task sequencing from local motion planning while incorporating realistic joint-space motion costs into the routing objective and can be solved using Google's OR tools [20].

1) Analytical Inverse Kinematics: Standard MoveIt-compatible solvers such as KDL, TracIK, and PICK-IK treat the linear actuator as an additional joint in an over-actuated chain and return a single, seed-dependent solution per call. For our 7-DoF system this is insufficient: different gantry positions give rise to qualitatively different arm configurations, and a single iterative call cannot enumerate them. We therefore implement a custom closed-form solver that exploits the UR16e's spherical wrist to decouple the position and orientation subproblems, and explicitly sweeps candidate gantry positions to cover the full workspace.

The 7-DoF system (6-DoF UR16e plus 1-DoF linear gantry) admits a closed-form IK solution for the arm at each fixed gantry position. We exploit the UR16e's spherical wrist to decompose the problem into five sequential subproblems: two shoulder-pan solutions from the horizontal wrist-center position, two wrist-2 solutions from the z -axis rotation constraint, and two elbow configurations from the resulting

planar 2R inverse kinematics, yielding up to eight analytical solutions per gantry position. To cover the extended workspace, the analytical solver is evaluated at up to 21 candidate gantry positions, offset from the current position in 0.1 m increments across ± 1.0 m and clamped to the actuator travel limits. Solutions are filtered by joint limits, a minimum end-effector clearance of 0.65 m from the gantry plate, and a triangle-area validity check on the wrist-forearm-gantry triangle to reject near-singular configurations. This procedure typically generates 50–100 valid IK candidates per target pose.

2) Multi-Objective IK Seed Selection: To select the best IK solution for the motion planner, we define a composite cost function over three normalized objectives. Let q_c denote the current joint configuration and q_k a candidate IK solution. The raw cost components are:

- *Joint distance:* weighted Euclidean distance $d_J(q_c, q_k) = \sqrt{\sum_{i=1}^7 w_i (q_{c,i} - q_{k,i})^2}$, where $w_1 = 0.1$ (gantry) and $w_{2-7} = 1.0$ (arm joints), penalizing large joint-space motions while permitting gantry travel.
- *Proximity penalty:* $p_P = 1/d_{ee}$, where d_{ee} is the Euclidean distance between the wrist and gantry plate links, discouraging configurations where the arm operates close to the gantry.
- *Area penalty:* $p_A = 1/A_\Delta$, where A_Δ is the area of the triangle formed by the gantry plate, wrist, and forearm links. Small triangle areas correspond to near-singular, collapsed arm configurations and serve as a geometric proxy for manipulability.

Each component is independently normalized to $[0, 1]$ across all valid candidates and combined as

$$C(q_k) = \alpha \hat{d}_J + \beta \hat{p}_P + \gamma \hat{p}_A,$$

where $\alpha, \beta, \gamma \geq 0$ are user-configurable weights exposed as runtime parameters. The candidate with the lowest $C(q_k)$ is used as the IK seed for the sampling-based motion planner.

We additionally compute the Yoshikawa manipulability index $\mu = \sqrt{\det(JJ^T)}$, where $J \in \mathbb{R}^{6 \times 7}$ is the geometric Jacobian, for each planned configuration. While μ is not directly incorporated into the cost function, it is recorded as a diagnostic metric to quantify the dexterity of the selected configuration. We observe empirically that the area penalty p_A correlates with manipulability, as collapsed arm triangles coincide with near-singular Jacobians.

3) Corridor Constraint: To produce predictable, human-safe motions in the collaborative workspace, we constrain the end-effector path to lie within a rectangular corridor between the start and goal poses. The corridor is defined as a 3D box whose long axis is aligned with the start-to-goal direction vector, with length equal to the segment distance plus a padding of 2×0.25 m, and a square cross-section of 2×0.5 m per side. The box is centered at the midpoint between start and goal and enforced as a MoveIt position constraint on the end-effector link.

TABLE III

RRT VS. CORRIDOR PLANNING COMPARISON (AVG. OVER SUCCESSFUL PLANS).

Strategy	SR (%) $\uparrow$	Plan (s) $\downarrow$	Exec (s) $\downarrow$	Path (m) $\downarrow$	Eff. $\uparrow$
Pure RRT	99.1	0.09	3.64	1.47	0.823
Corridor	93.4	29.87	6.97	1.26	0.910

B. Vision

C. Agentic AI

IV. RESULTS

WE NEED SOME RESULTS ON THE PLANNER. FOR EXAMPLE, WE COULD PLOT THE OVERALL PATH LENGTH AS A FUNCTION OF $N=1,2,3\dots k$ IK SOLUTIONS.

We have measured accuracy for true-positive detection and intersection-over-unions (IoU) for X objects in the battery. We obtain $XX\%$ accuracy and $XX\%$ IoU on average. We can detect screws with accuracy XX , busbars with YY ...

We have also timed a disassembly sequence for the top cover, by performing disassembly $N=5$ times.

A. Planning

We evaluate the motion planner in two studies using 1,000 randomly sampled goal poses within the battery workspace, with a planning timeout of 20 s. All metrics are averaged over successful plans only.

1) *RRT vs. Corridor-Constrained Planning*: We first compare unconstrained RRT planning against corridor-constrained planning (Section III-A) using 1,000 goals at a fixed height of $z = 1.31$ m. Table III summarizes the results. RRT achieves a higher success rate (99.1% vs. 93.4%) with dramatically lower planning times (0.09 s vs. 29.87 s on average). However, the corridor constraint produces more direct paths, with a path efficiency of 0.910 compared to 0.823 for unconstrained RRT.

Figure 4 shows the plan time distributions on a logarithmic scale. RRT plan times cluster tightly around 0.10 s, while corridor-constrained planning exhibits a broad distribution with a median of 13.75 s, with many plans approaching or exceeding the 20 s timeout.

Figure 5 shows a per-goal comparison of plan times. Nearly all points lie above the diagonal, confirming that the corridor constraint increases planning time for every goal. Blue points indicate goals where only RRT succeeded, predominantly at high corridor plan times where the constrained planner timed out.

Figure 6 shows the spatial distribution of failures. RRT failures (9 total) are scattered, while corridor failures (66 total) cluster at the positive- x workspace boundary, where the corridor box extends beyond the reachable workspace.

2) *Cost Function Ablation Study*: We evaluate the effect of the three IK seed selection cost components (joint distance, proximity penalty, area penalty) using eight weight configurations in an ablation study. Goals are sampled with

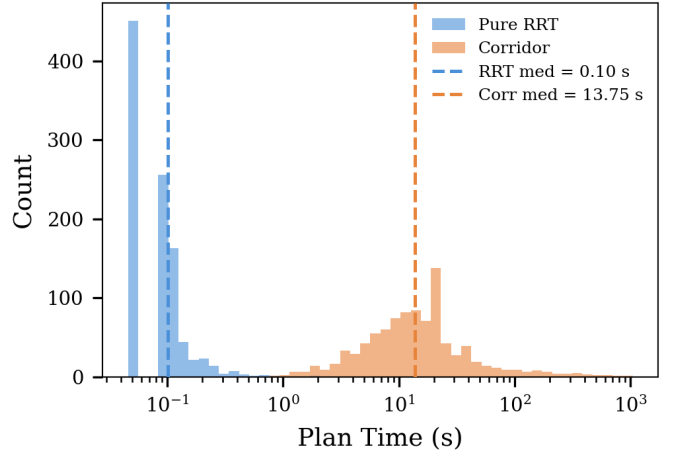


Fig. 4. Plan time distributions for unconstrained RRT vs. corridor-constrained planning. Dashed line indicates the 20 s timeout.

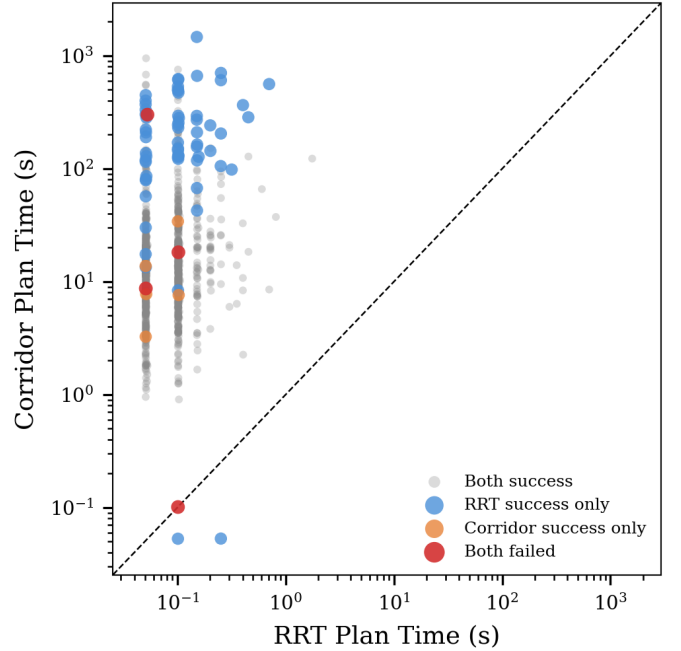


Fig. 5. Per-goal plan time comparison. Each point represents one of 1,000 goals. Points above the diagonal indicate corridor planning was slower.

varying heights ($z \in [1.0, 1.47]$ m) and planned from a fixed home position. Table IV summarizes the results.

The baseline configuration ($\alpha = \beta = \gamma = 0$, uniform random IK seed selection) achieves the highest success rate (99.6%) but the slowest average plan time (0.96 s) and longest paths (2.40 m). The equally-weighted configuration ($\alpha = \beta = \gamma = 1$) reduces plan time to 0.22 s while maintaining a 99.3% success rate and achieving high manipulability (0.628). Removing the proximity penalty ($\sim$ Prox) yields the fastest execution time (3.98 s) and shortest paths (1.52 m), but reduces the success rate to 97.9% and manipulability to 0.424. The proximity-only configuration achieves the highest manipulability (0.700), as configurations that maintain clearance from the gantry tend to be more dexterous. Figure 7

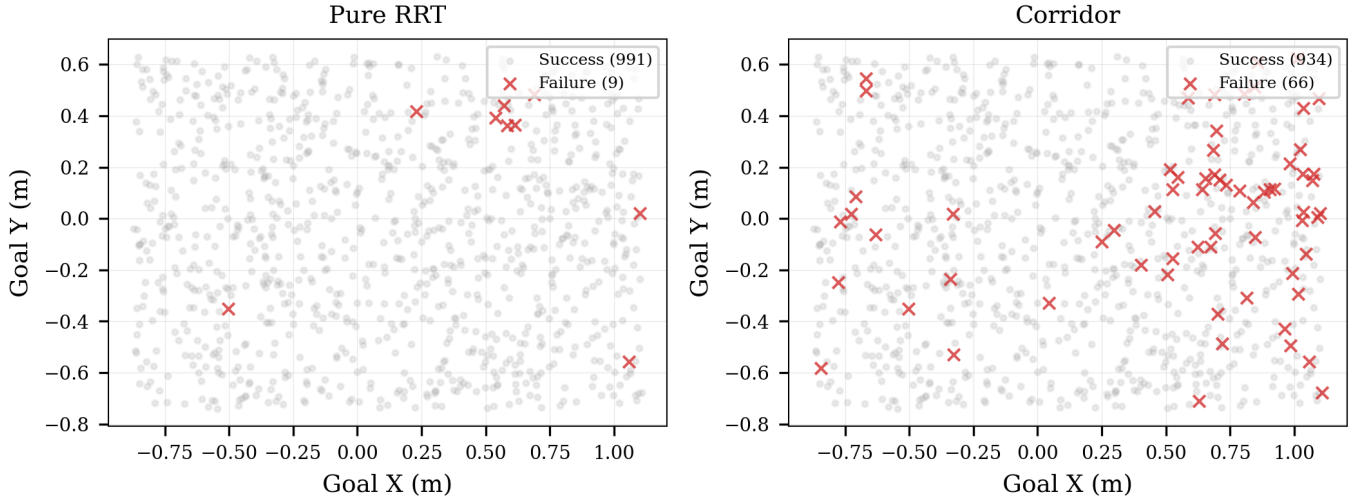


Fig. 6. Spatial distribution of planning failures. Left: unconstrained RRT (9 failures). Right: corridor-constrained planning (66 failures), with failures concentrated at the workspace boundary.

TABLE IV
COST FUNCTION ABLATION RESULTS (AVG. OVER SUCCESSFUL PLANS).

BOLD = BEST, UNDERLINE = SECOND BEST.

Config.	SR (%) $\uparrow$	Plan (s) $\downarrow$	Exec (s) $\downarrow$	Path (m) $\downarrow$	Manip. $\uparrow$
Base	99.6	0.96	6.24	2.40	0.481
Area	99.0	0.53	6.28	2.52	0.421
Joint	98.3	0.56	4.53	<u>1.64</u>	0.489
Prox	99.3	0.38	4.24	1.87	0.700
Area	<u>99.4</u>	0.48	4.33	1.78	<u>0.699</u>
Joint	99.1	<u>0.27</u>	4.26	2.01	0.642
Prox	97.9	<u>0.27</u>	3.98	1.52	0.424
Equal	99.3	0.22	<u>4.10</u>	1.70	0.628

illustrates this relationship: compared to the random-seed baseline, the proximity-only configuration consistently selects configurations with higher manipulability scores across all goal distances, with the largest improvement at close-range goals where multiple arm configurations compete.

Figure 8 shows the per-goal distributions across all eight configurations. The plan time panel (a) reveals that the baseline has a significantly heavier tail compared to the cost-informed configurations. The manipulability panel (d) shows a clear separation: configurations that include the proximity term (Prox, ~Area, ~Joint, Equal) consistently achieve higher manipulability scores than those without it.

Figure 9 maps the spatial distribution of planning failures across all eight configurations. Goals that failed under multiple configurations cluster near the positive- x workspace boundary, corresponding to extended-reach poses where the arm approaches kinematic limits.

B. Vision

C. Agentic AI

V. DISCUSSION

As in [8], we observe robots to work significantly slower than human workers. Similar to [8], we operate the robots

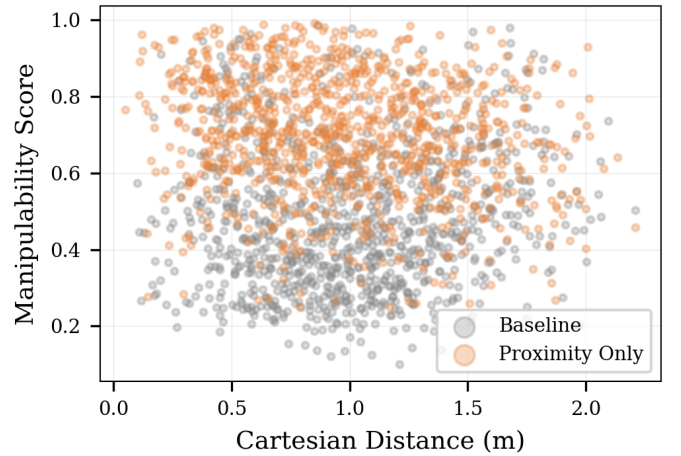


Fig. 7. Manipulability score vs. Cartesian goal distance for the baseline (grey) and proximity-only (orange) configurations. Each point represents one successful plan.

far below their maximum speed, but we note that a robot does not necessarily need to be faster than a human to save labor. In this case study, we can reduce the time required to remove all the top-cover screws from 501 seconds with one worker to 400 seconds when the worker is supported by a robot.

A. Planning

B. Vision

C. Agentic AI

VI. CONCLUSIONS

We have demonstrated how the agentic AI can both control the robot directly (identifying and disassembling nuts) as well as soliciting human workers for removing screws and the top cover. In the future, we will extend this system to perform a complete disassembly sequence, which will also require humans to change nut-runner bits on the agent's

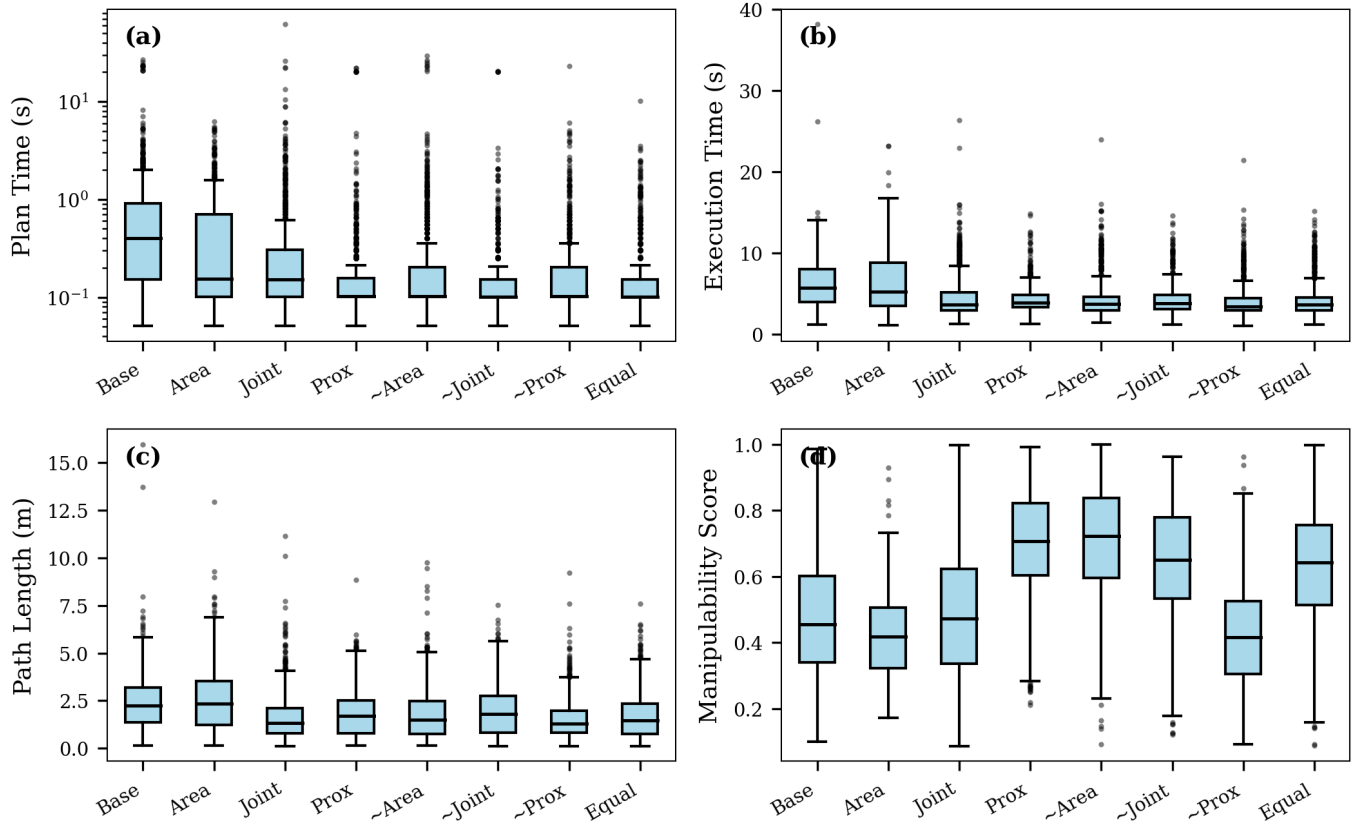


Fig. 8. Per-goal distributions across eight cost function configurations. (a) Plan time (log scale), (b) execution time, (c) path length, (d) manipulability score. All metrics computed over successful plans only.

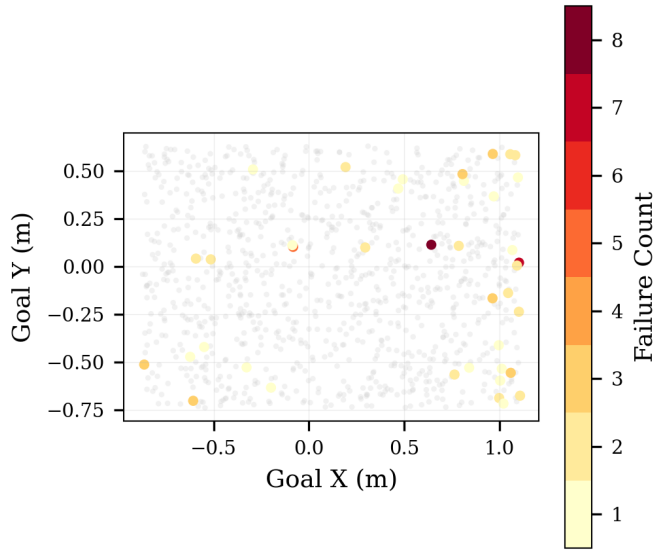


Fig. 9. Goal difficulty heatmap. Each of 1,000 goals is colored by the number of cost configurations (out of 8) that failed to find a plan.

request. In addition to performing a timing analysis and comparing this with manual assembly, we are interested in the human-robot interaction aspects. Here, the proposed solution offers both potential for labor savings as well as consistent instructions, which we wish to quantify by a

survey.

Finally, we are interested in extending the proposed solution by an additional mobile manipulator that can perform tasks such as removing screws, changing tools, and transporting small items.

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APPENDIX

Appendices should appear before the acknowledgment.

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